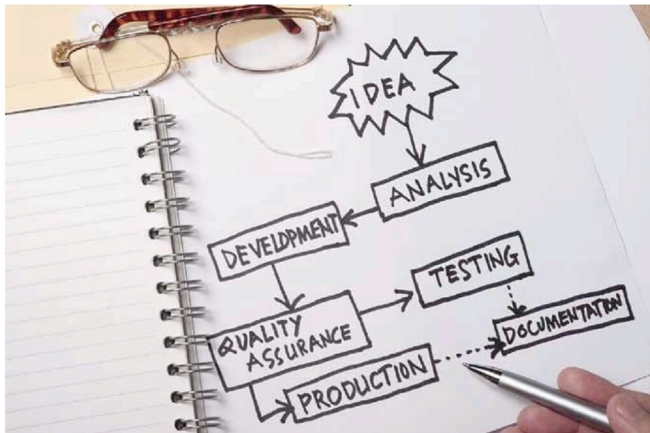


A Peek Into the CUnit Testing Framework

CUnit is a free and open source, unit testing, software framework written in C. It provides a very simple interface to write unit tests, and has a number of assertions for testing data and functions.



CUnit is created as a library that links with the user's code under test. It is packaged and is available in most GNU/Linux distributions. It has been released under the LGPLv2+ licence. You can install it on Fedora, for example, using the following command:

```
{.bash}
$ sudo yum install CUnit
```

The CUnit framework helps manage test cases, the test registry and test suites. A systematic approach is to write unit tests for your code, initialise the CUnit test registry, add the test suites to the test registry, and then add the unit tests to the suites. Set-up and clean-up functions can be written for each test suite. All the test suites can be executed together or can be run selectively. The user interface method to run the tests needs to be decided before executing them. The test registry needs to be cleaned up before returning the error condition and exiting from the test run. Consider the

following simple example:

```
{.c}
#include <CUnit/CUnit.h>

int init_suite(void) { return 0; }
int clean_suite(void) { return 0; }

int
is_even (int x)
{
    return (x % 2 == 0);
}

void
test_is_even (void)
{
    CU_ASSERT(is_even(1) == 0);
    CU_ASSERT(is_even(2) == 1);
}
```